Principal Component Analysis (PCA) Report

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1. Introduction

1.1 Objective

This report demonstrates Principal Component Analysis (PCA) on a custom dataset containing student performance metrics across different subjects and activities.

1.2 About the Dataset

The dataset contains information about 100 students with the following variables: - Math_Score - Science_Score - English_Score - Sports_Score - Arts_Score - Attendance_Rate - Study_Hours - Extracurricular_Hours

2. Loading Required Packages

```
# Install packages if not already installed
if(!require(dplyr)) install.packages('dplyr')

## Loading required package: dplyr

## Warning: package 'dplyr' was built under R version 4.4.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union

if(!require(factoextra)) install.packages('factoextra')

## Loading required package: factoextra

## Warning: package 'factoextra' was built under R version 4.4.3
```

```
## Loading required package: ggplot2

## Warning: package 'ggplot2' was built under R version 4.4.3

## Welcome! Want to learn more? See two factoextra-related books at
https://goo.gl/ve3WBa
```

```r
if(!require(ggplot2)) install.packages('ggplot2')
if(!require(corrplot)) install.packages('corrplot')
```

```
## Loading required package: corrplot

## Warning: package 'corrplot' was built under R version 4.4.3

## corrplot 0.95 loaded
```

```r
# Load libraries
library(dplyr)
library(factoextra)
library(ggplot2)
library(corrplot)
```

3. Data Preparation

3.1 Creating Custom Dataset

```r
# Set seed for reproducibility
set.seed(123)

# Create custom dataset
n <- 100  # number of students

student_data <- data.frame(
  Student_ID = paste0("S", 1:n),
  Math_Score = round(rnorm(n, mean = 75, sd = 12)),
  Science_Score = round(rnorm(n, mean = 72, sd = 13)),
  English_Score = round(rnorm(n, mean = 78, sd = 11)),
  Sports_Score = round(rnorm(n, mean = 68, sd = 15)),
  Arts_Score = round(rnorm(n, mean = 70, sd = 14)),
  Attendance_Rate = round(runif(n, min = 65, max = 98), 1),
  Study_Hours = round(rnorm(n, mean = 5, sd = 1.5), 1),
  Extracurricular_Hours = round(rnorm(n, mean = 3, sd = 1), 1)
)

# Ensure scores are within valid range (0-100)
student_data <- student_data %>%
  mutate(across(Math_Score:Arts_Score, ~pmin(pmax(., 0), 100)))

# Display first few rows
head(student_data, 10)
```

```
##    Student_ID Math_Score Science_Score English_Score Sports_Score
Arts_Score
## 1          S1         68            63           100           57
69
## 2          S2         72            75            92           57
54
## 3          S3         94            69            75           54
61
## 4          S4         76            67            84           52
70
## 5          S5         77            60            73           61
79
## 6          S6         96            71            73           73
47
## 7          S7         81            62            69           38
65
## 8          S8         60            50            71           71
81
## 9          S9         67            67            96           87
62
## 10        S10         70            84            77           99
73
##    Attendance_Rate Study_Hours Extracurricular_Hours
## 1             74.0         6.3                   1.1
## 2             84.6         3.8                   3.1
## 3             70.3         6.7                   3.6
## 4             93.2         5.4                   1.5
## 5             93.0         7.5                   3.5
## 6             80.8         2.8                   2.2
## 7             90.5         4.9                   4.0
## 8             74.7         4.2                   3.5
## 9             67.2         4.7                   3.8
## 10            79.5         4.1                   3.1
```

3.2 Data Summary

```r
# Summary statistics
summary(student_data[, -1])  # Exclude Student_ID
```

```
##    Math_Score     Science_Score    English_Score     Sports_Score
##  Min.   : 47.00   Min.   : 45.00   Min.   : 59.00   Min.   : 31.00
##  1st Qu.: 69.00   1st Qu.: 61.75   1st Qu.: 72.00   1st Qu.: 57.00
##  Median : 76.00   Median : 69.00   Median : 78.50   Median : 68.00
##  Mean   : 76.03   Mean   : 70.49   Mean   : 79.29   Mean   : 67.34
##  3rd Qu.: 83.00   3rd Qu.: 78.25   3rd Qu.: 86.25   3rd Qu.: 78.25
##  Max.   :100.00   Max.   :100.00   Max.   :100.00   Max.   :100.00
##    Arts_Score     Attendance_Rate  Study_Hours    Extracurricular_Hours
##  Min.   : 33.00   Min.   :65.40   Min.   :0.800   Min.   :0.800
##  1st Qu.: 64.75   1st Qu.:74.00   1st Qu.:3.700   1st Qu.:2.300
##  Median : 72.00   Median :80.35   Median :4.800   Median :3.100
```

```
##  Mean   : 71.45   Mean   :80.81   Mean   :4.796   Mean   :3.041
##  3rd Qu.: 80.00   3rd Qu.:88.53   3rd Qu.:5.900   3rd Qu.:3.700
##  Max.   :100.00   Max.   :97.80   Max.   :8.100   Max.   :5.700
```

3.3 Data Structure

```r
str(student_data)
```

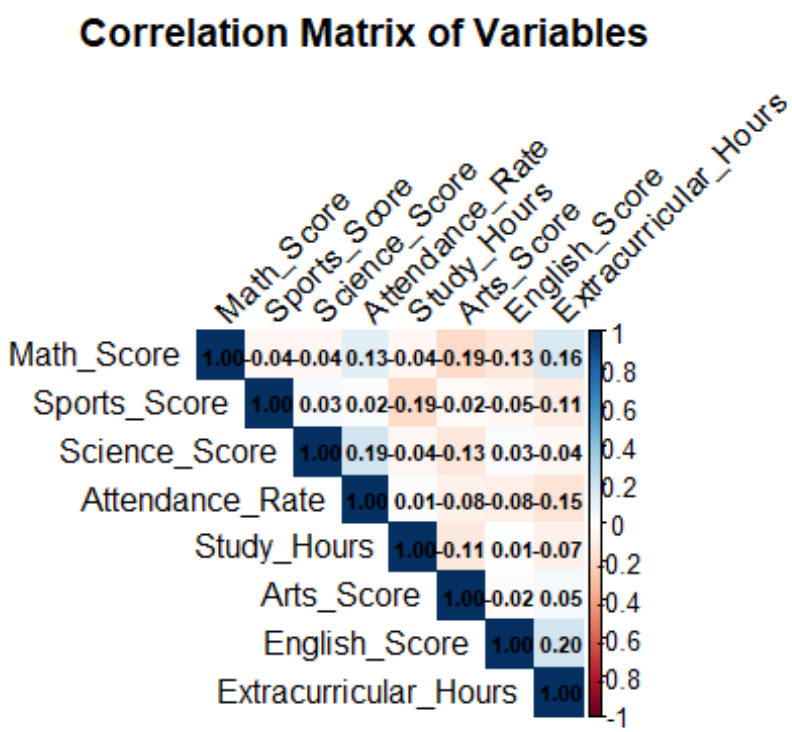
```
## 'data.frame':    100 obs. of  9 variables:
##  $ Student_ID           : chr  "S1" "S2" "S3" "S4" ...
##  $ Math_Score           : num  68 72 94 76 77 96 81 60 67 70 ...
##  $ Science_Score        : num  63 75 69 67 60 71 62 50 67 84 ...
##  $ English_Score        : num  100 92 75 84 73 73 69 71 96 77 ...
##  $ Sports_Score         : num  57 57 54 52 61 73 38 71 87 99 ...
##  $ Arts_Score           : num  69 54 61 70 79 47 65 81 62 73 ...
##  $ Attendance_Rate      : num  74 84.6 70.3 93.2 93 80.8 90.5 74.7 67.2
79.5 ...
##  $ Study_Hours          : num  6.3 3.8 6.7 5.4 7.5 2.8 4.9 4.2 4.7 4.1 ...
##  $ Extracurricular_Hours: num  1.1 3.1 3.6 1.5 3.5 2.2 4 3.5 3.8 3.1 ...
```

4. Exploratory Data Analysis

4.1 Correlation Matrix

```r
# Calculate correlation matrix (excluding Student_ID)
cor_matrix <- cor(student_data[, -1])

# Visualize correlation matrix
corrplot(cor_matrix, method = "color", type = "upper",
         order = "hclust", addCoef.col = "black",
         tl.col = "black", tl.srt = 45,
         number.cex = 0.7,
         title = "Correlation Matrix of Variables",
         mar = c(0,0,2,0))
```



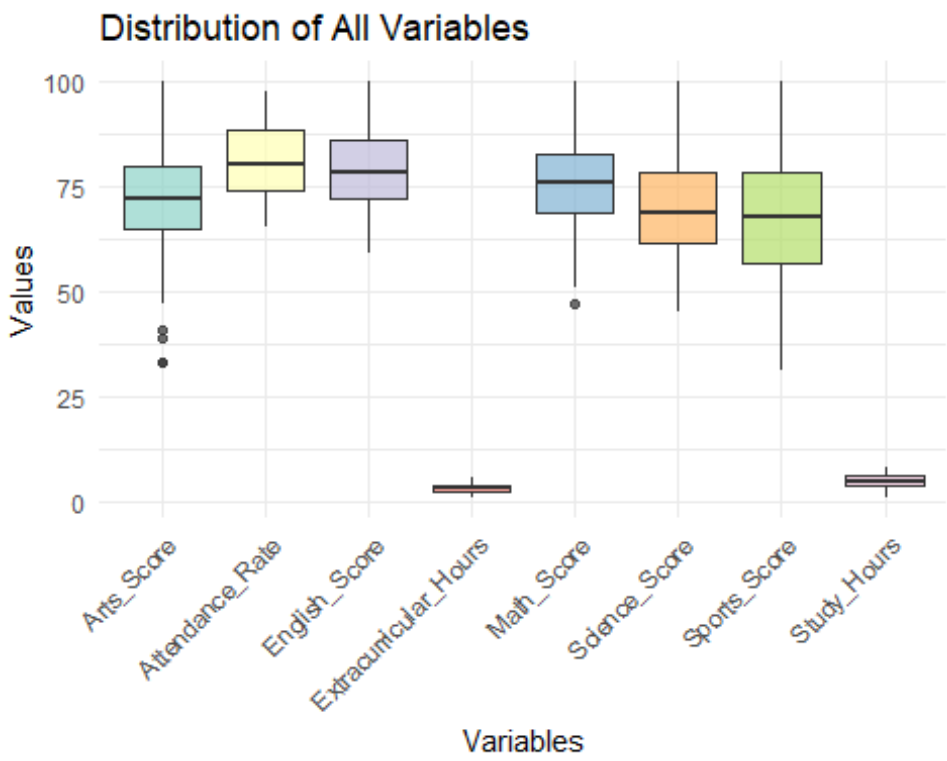
4.2 Distribution of Variables

```
# Reshape data for plotting
library(tidyr)

## Warning: package 'tidyr' was built under R version 4.4.3

data_long <- student_data[, -1] %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to =
"Value")

# Create boxplot
ggplot(data_long, aes(x = Variable, y = Value, fill = Variable)) +
  geom_boxplot(alpha = 0.7) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none") +
  labs(title = "Distribution of All Variables",
       x = "Variables",
       y = "Values") +
  scale_fill_brewer(palette = "Set3")
```



5. Principal Component Analysis

5.1 Preparing Data for PCA

```
# Select numeric columns (exclude Student_ID)
pca_data <- student_data[, -1]

# Check for missing values
sum(is.na(pca_data))

## [1] 0
```

5.2 Performing PCA with Scaling

Note: We use `scale = TRUE` to standardize variables since they have different scales.

```
# Perform PCA
pc <- prcomp(pca_data, scale = TRUE)

# Display PCA results
print(pc)

## Standard deviations (1, .., p=8):
## [1] 1.1892387 1.1262659 1.0863954 1.0569981 0.9766556 0.8896839 0.8387099
## [8] 0.7555849
##
```

```
## Rotation (n x k) = (8 x 8):
##                               PC1         PC2         PC3         PC4
## Math_Score             0.24639134 -0.52893190 -0.40011439 -0.38977039
## Science_Score          0.37690730 -0.01218283 -0.07819316  0.62603850
## English_Score         -0.36397548 -0.19392801 -0.02157804  0.63558287
## Sports_Score           0.17720072  0.46182680 -0.41185260  0.06216476
## Arts_Score            -0.39775433  0.41008811 -0.01342595 -0.14347014
## Attendance_Rate        0.55242044 -0.05220961 -0.02033071  0.12378787
## Study_Hours            0.05006305 -0.29138976  0.70949487 -0.03496233
## Extracurricular_Hours -0.40923196 -0.46193556 -0.39966182  0.10580794
##                               PC5         PC6         PC7         PC8
## Math_Score             0.01135638 -0.09354440 -0.03907701 -0.58060601
## Science_Score          0.23885755  0.53505721 -0.21835470 -0.26217584
## English_Score         -0.15945549 -0.49525967  0.21785665 -0.32758643
## Sports_Score          -0.51424490 -0.24391933 -0.50623557 -0.04012452
## Arts_Score             0.62584583 -0.13776392 -0.31094424 -0.38145389
## Attendance_Rate        0.46231998 -0.59926653 -0.02880937  0.32043553
## Study_Hours           -0.16584636 -0.12455844 -0.59164806 -0.12211934
## Extracurricular_Hours  0.14147950  0.08105788 -0.44664809  0.47131572
```

5.3 PCA Summary

```r
# Summary of PCA
summary(pc)
```

```
## Importance of components:
##                           PC1    PC2    PC3    PC4    PC5     PC6     PC7
## Standard deviation     1.1892 1.1263 1.0864 1.0570 0.9767 0.88968 0.83871
## Proportion of Variance 0.1768 0.1586 0.1475 0.1397 0.1192 0.09894 0.08793
## Cumulative Proportion  0.1768 0.3353 0.4829 0.6225 0.7418 0.84071 0.92864
##                            PC8
## Standard deviation     0.75558
## Proportion of Variance 0.07136
## Cumulative Proportion  1.00000
```

#**Proportion of Variance**: Shows the percentage of total variance explained by each component: - PC1 explains **17.68%** of the total variance - PC2 explains **15.86%** of the total variance - Together, the first two components explain **33.53%** of the variance - Each subsequent component explains progressively less variance

- **Cumulative Proportion**: Running total of variance explained:
  - First 3 components (PC1-PC3) explain **48.29%** of total variance
  - First 4 components (PC1-PC4) explain **62.25%** of total variance
  - First 5 components (PC1-PC5) explain **74.18%** of total variance
  - First 6 components explain **84.07%** - often sufficient for analysis

Key Insights:

1. **No Single Dominant Component**: PC1 captures only 17.68% of variance, suggesting the data has relatively evenly distributed variance across multiple dimensions rather than one dominant pattern.

2. **Gradual Decline**: The variance explained decreases gradually (17.68% → 15.86% → 14.75%), rather than a sharp drop, indicating multiple underlying factors contribute to the variation.

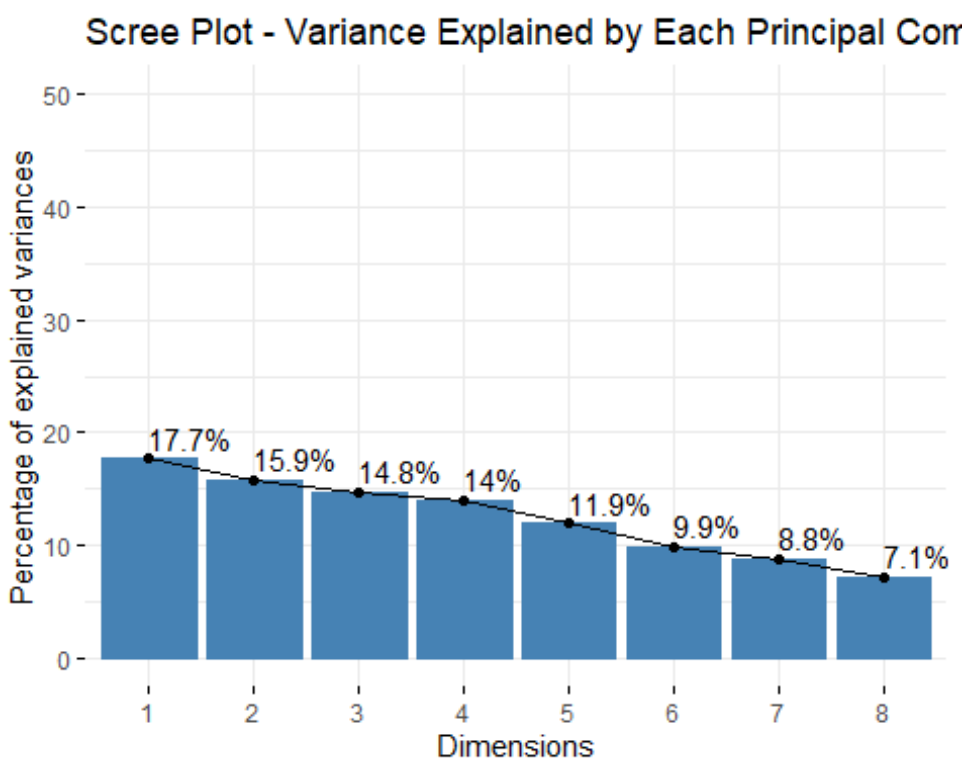
3. **Dimensionality Reduction**: To retain approximately 80% of information, we need at least **6 components** (84.07%), reducing from 8 original variables - a 25% reduction in dimensionality.

4. **Practical Recommendation**: Based on common thresholds:

    - For 70% variance: Use **5 components**
    - For 80% variance: Use **6 components**
    - For 90% variance: Use **7 components** # 6. PCA Visualization

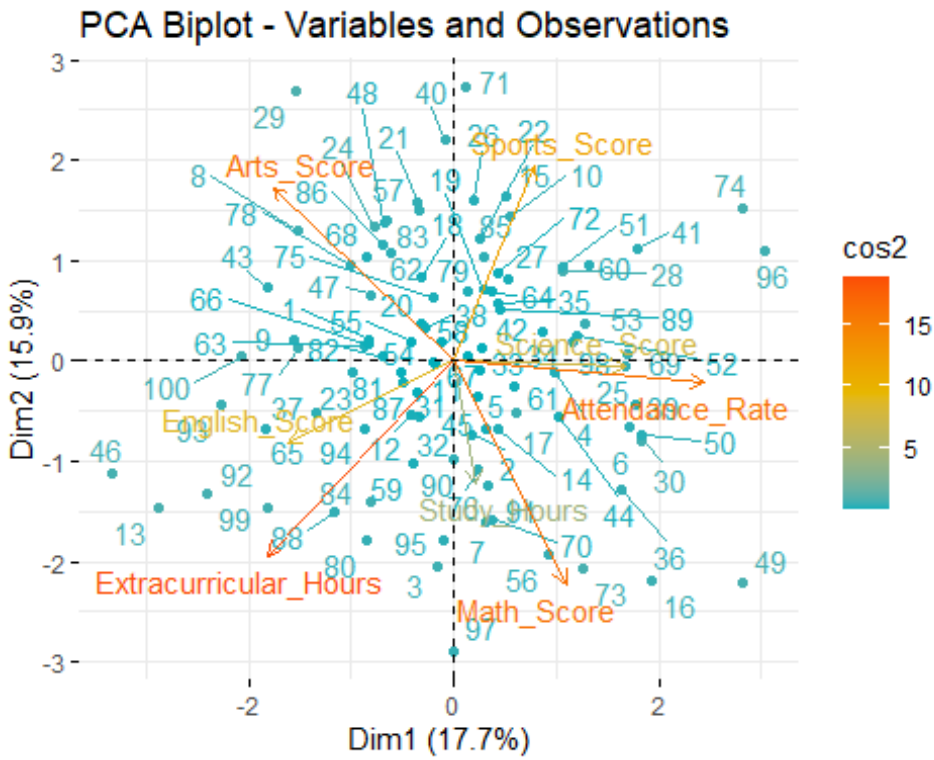
6.1 Scree Plot

```
# Scree plot showing variance explained by each component
fviz_eig(pc, addlabels = TRUE, ylim = c(0, 50),
         main = "Scree Plot - Variance Explained by Each Principal
Component",
         barfill = "steelblue", barcolor = "steelblue")
```



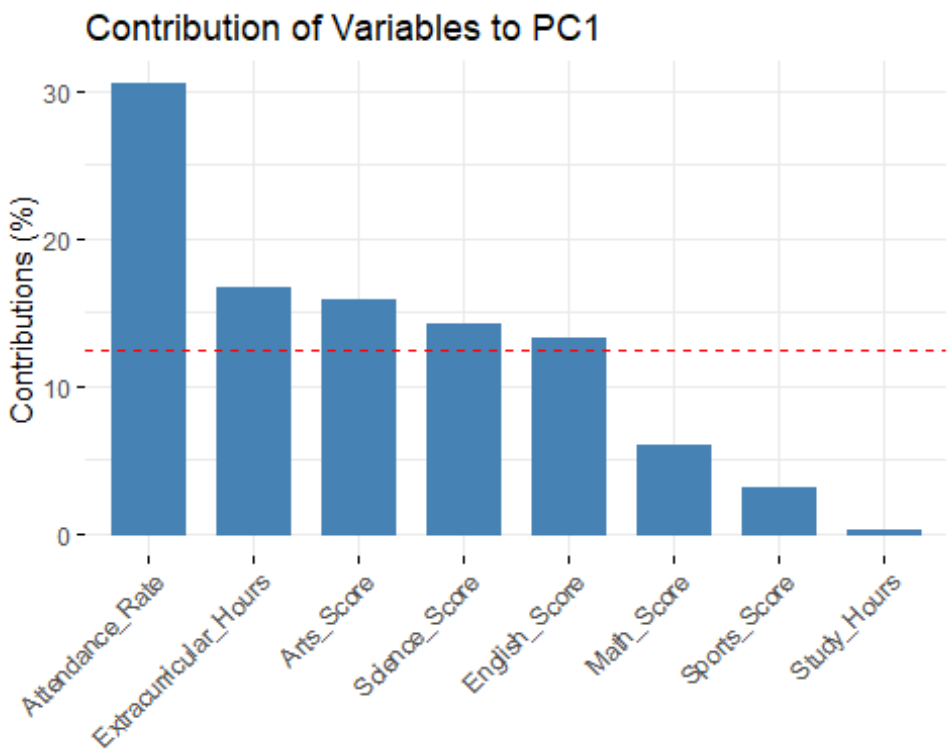
6.2 Biplot

```
# Biplot showing both observations and variables
fviz_pca_biplot(pc,
                col.ind = "cos2",
                gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
                col.var = "contrib",
                repel = TRUE,
                title = "PCA Biplot - Variables and Observations")
```

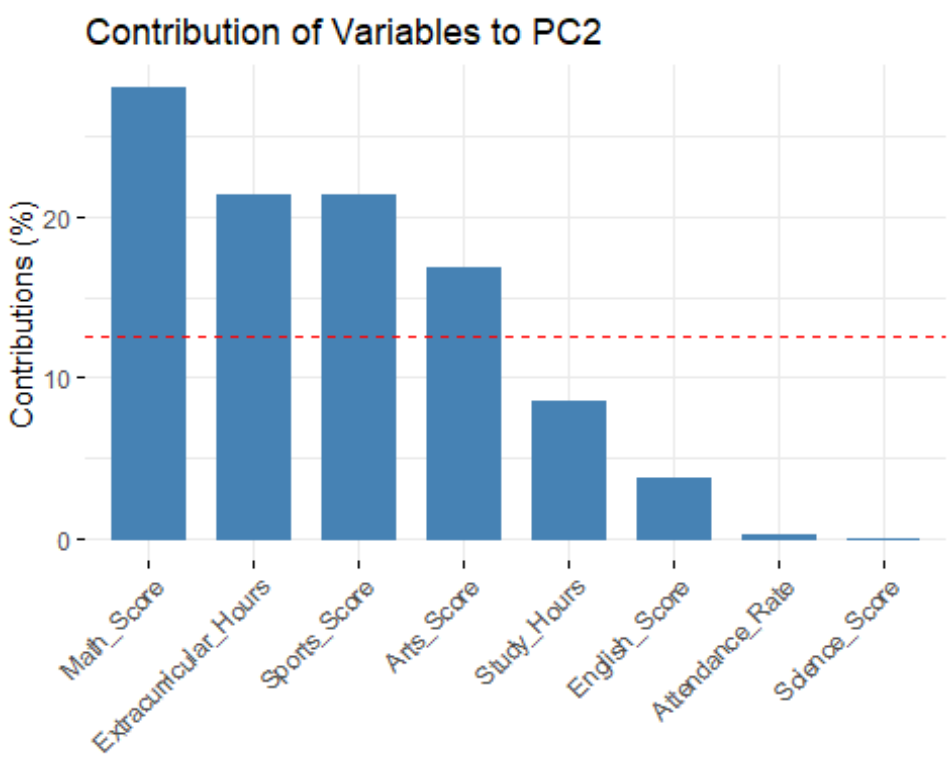


6.3 Variable Contribution Plot

```
# Contributions of variables to PC1
fviz_contrib(pc, choice = "var", axes = 1, top = 10,
             title = "Contribution of Variables to PC1")
```

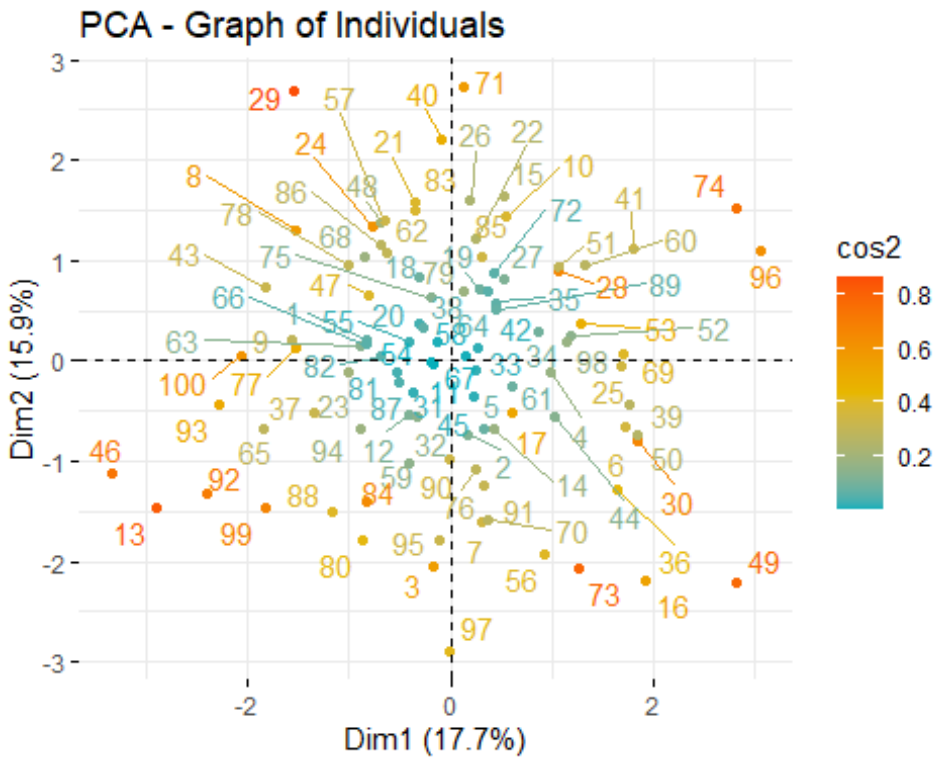



```r
# Contributions of variables to PC2
fviz_contrib(pc, choice = "var", axes = 2, top = 10,
             title = "Contribution of Variables to PC2")
```



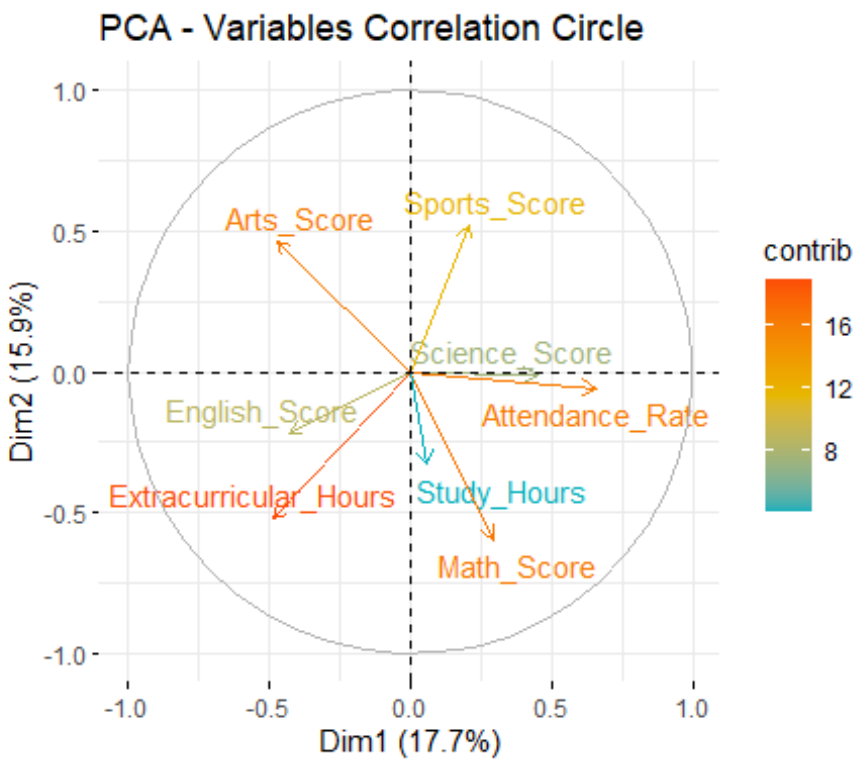
6.4 Individual Observations Plot

```
# Plot of individuals
fviz_pca_ind(pc,
             col.ind = "cos2",
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
             repel = TRUE,
             title = "PCA - Graph of Individuals")
```



6.5 Variables Correlation Circle

```
# Variables correlation circle
fviz_pca_var(pc,
             col.var = "contrib",
             gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"),
             repel = TRUE,
             title = "PCA - Variables Correlation Circle")
```


PCA - Variables Correlation Circle
Arts_Score
Sports_Score
Science_Score
English_Score
Attendance_Rate
Extracurricular_Hours
Study_Hours
Math_Score
contrib
16
12
8
1.0
0.5
0.0
-0.5
-1.0
Dim2 (15.9%)
Dim1 (17.7%)